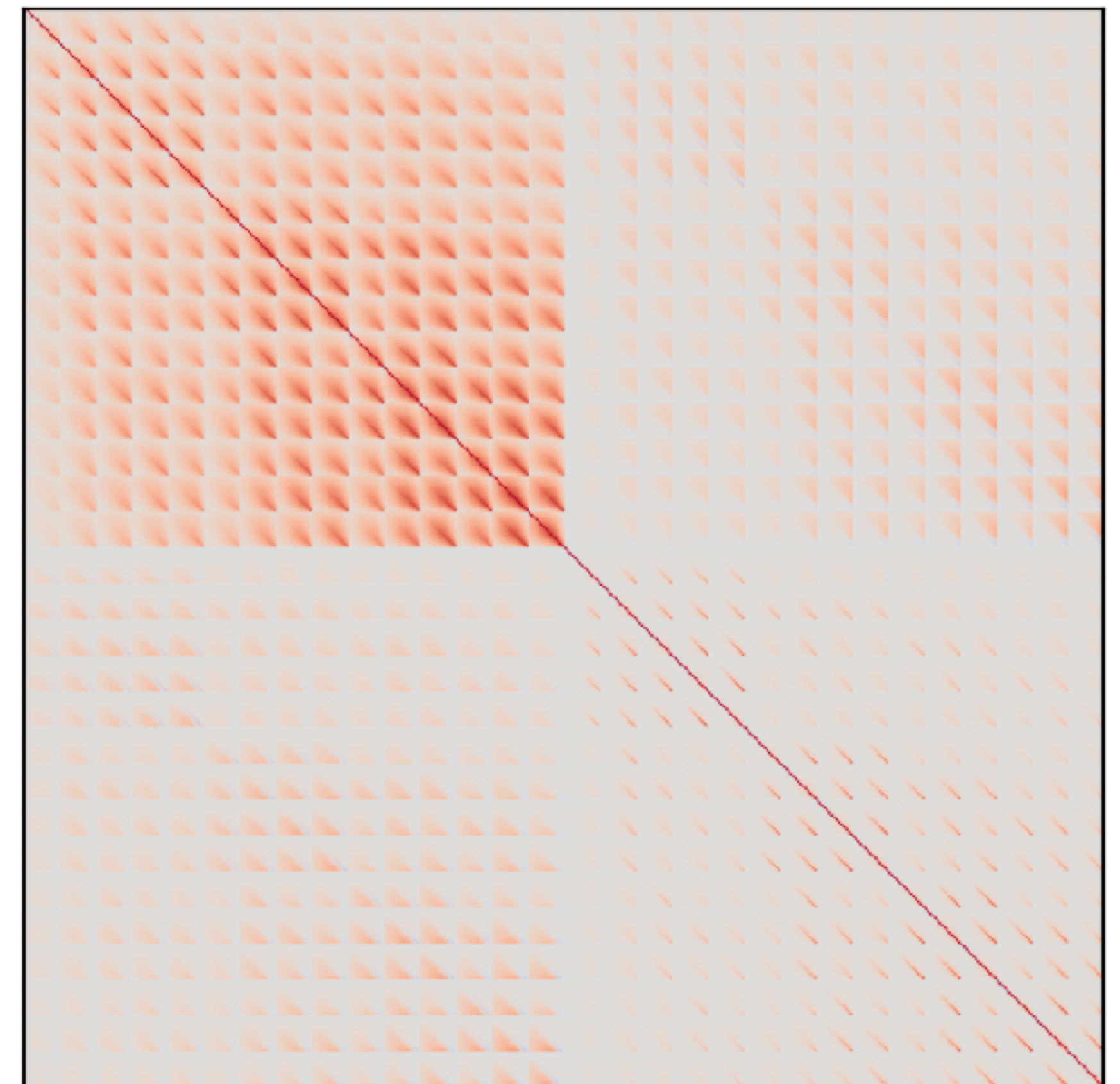


Covariance Matrix Estimation

Jed Homer LMU | ODSL



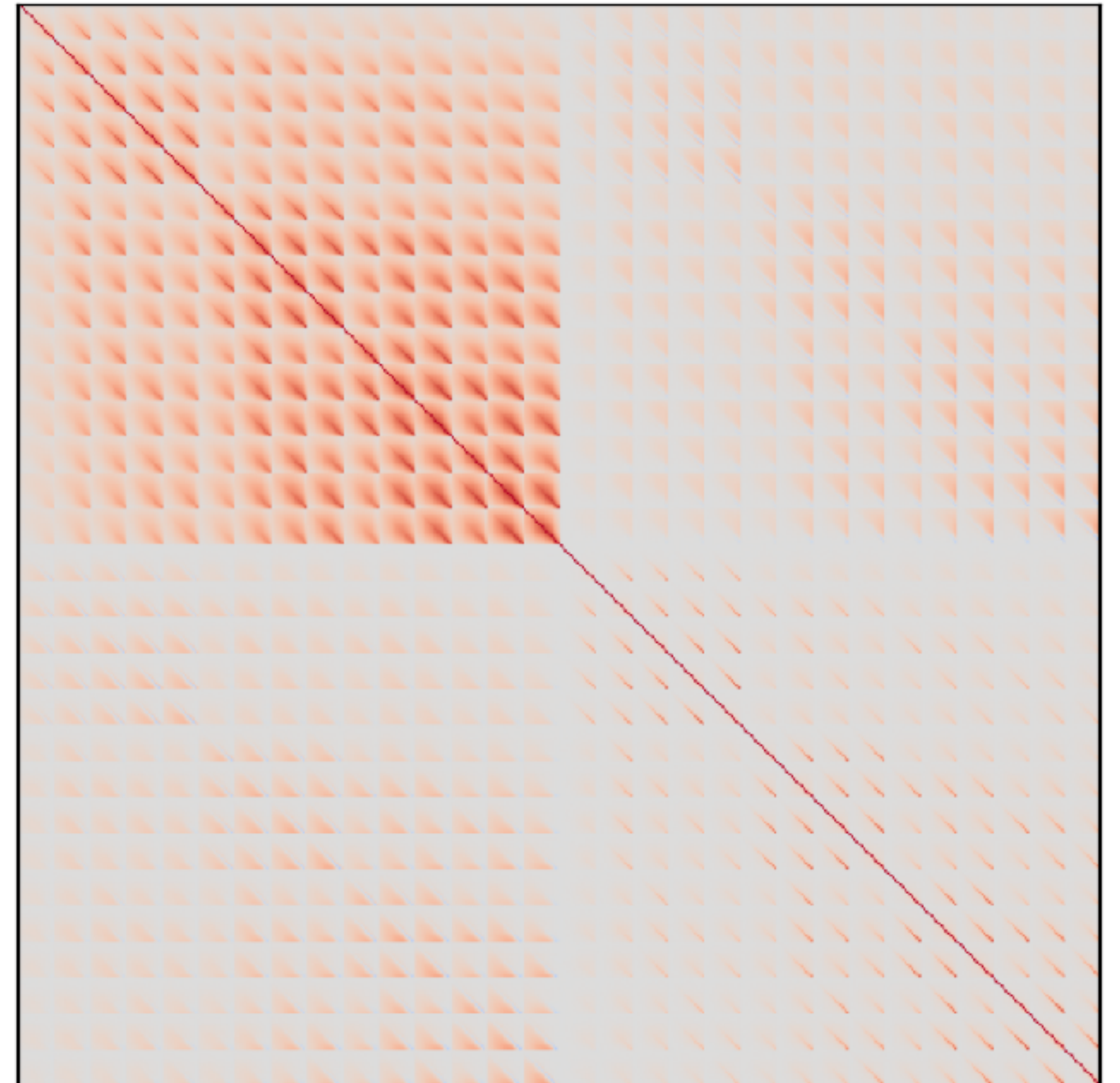
Σ



Synopsis

- What is a covariance matrix, and why is it so important?
- What happens to posteriors if we don't know it?
- If we can accurately simulate it, how do we account for noise in the estimate?

Σ



Likelihood PDF

- Inference requires a likelihood **PDF** which sets the limit of **information in $\hat{x}$ on π**

$$p(\hat{x} \mid x[\pi], \pi)$$

- Data $\hat{x}$ scatters around the expectation $x[\pi]$ as a function of π

Likelihood PDF

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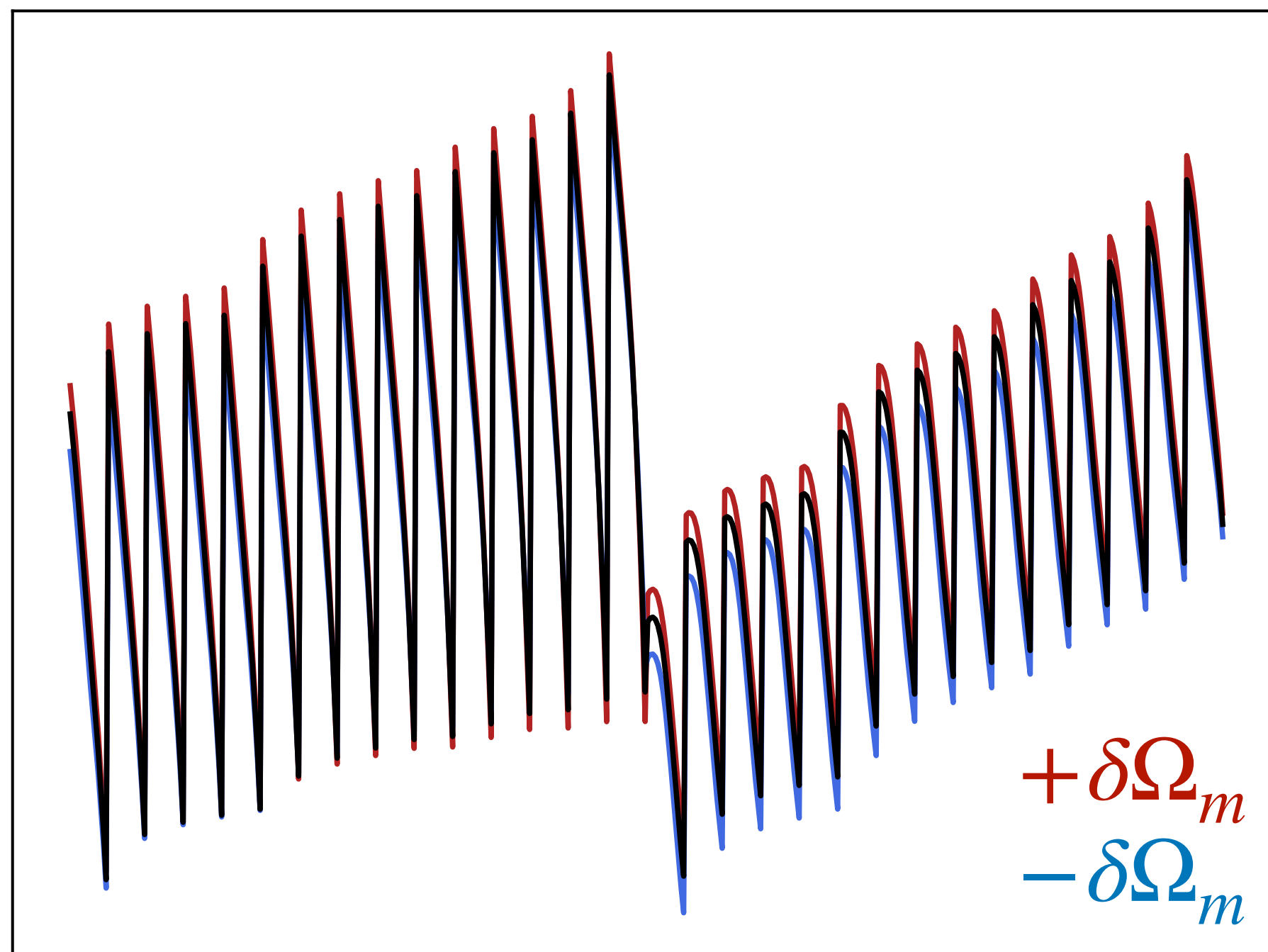
$$p(\hat{x} \mid x[\pi], \pi)$$

e.g. $\mathcal{G}[\hat{x} \mid x[\pi], \Sigma]$

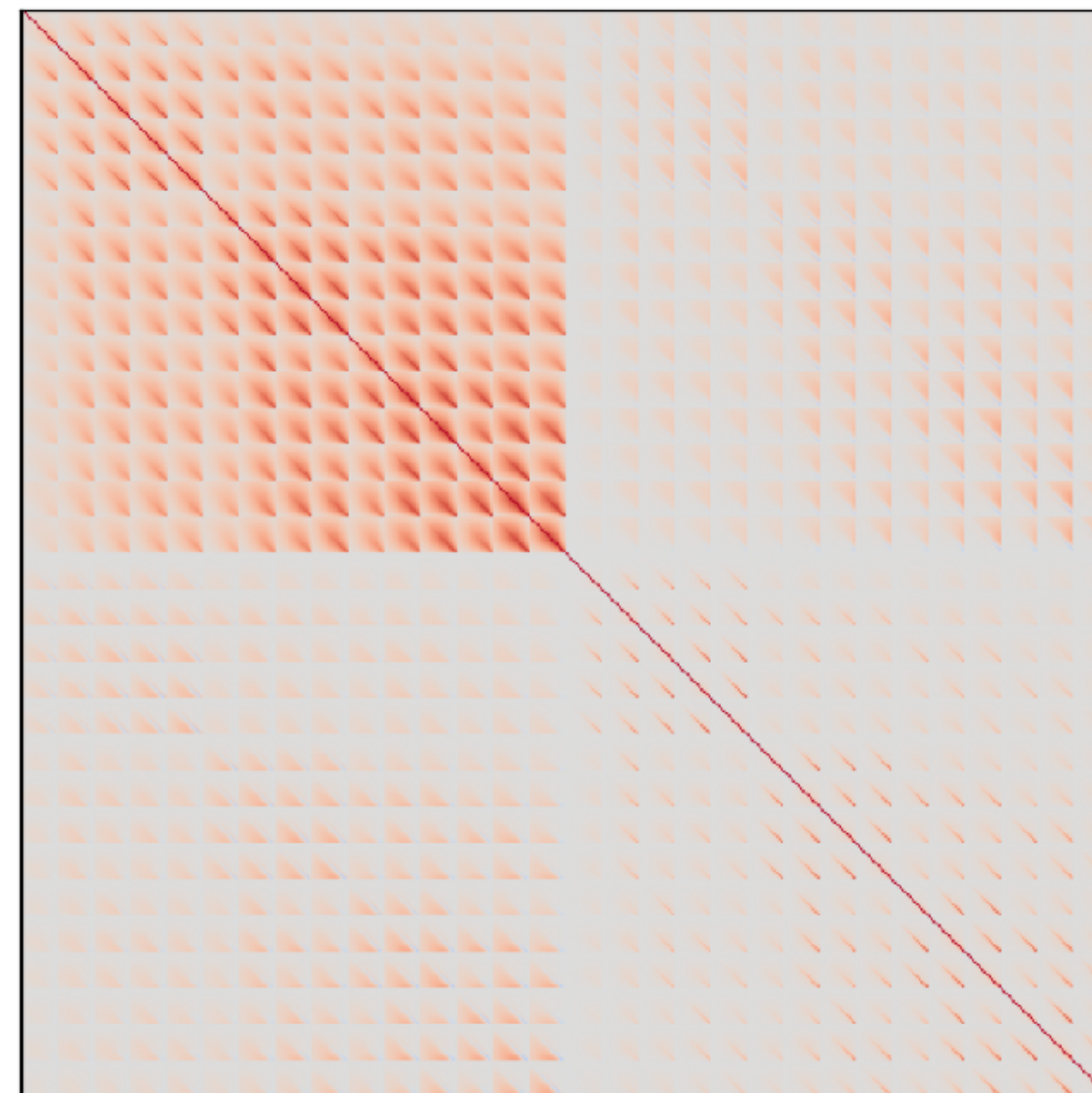
$$\propto \exp \left[-\frac{1}{2} (\hat{x} - x[\pi])^T \Sigma^{-1} (\hat{x} - x[\pi]) \right]$$

$$p(\hat{x} \mid \pi) = \mathcal{G} \left[\hat{x} \mid x[\pi], \Sigma \right]$$

$x[\pi]$

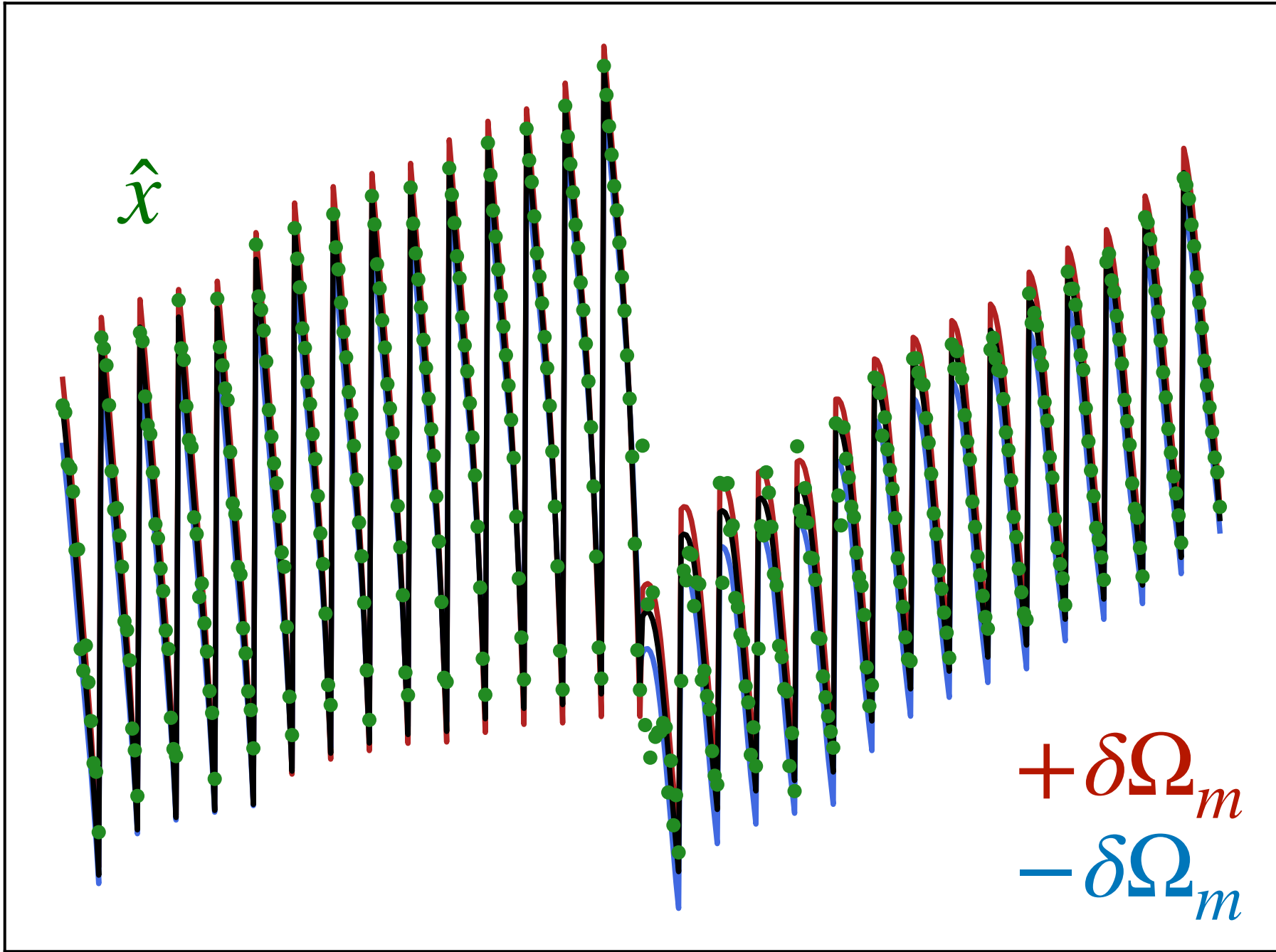


Σ

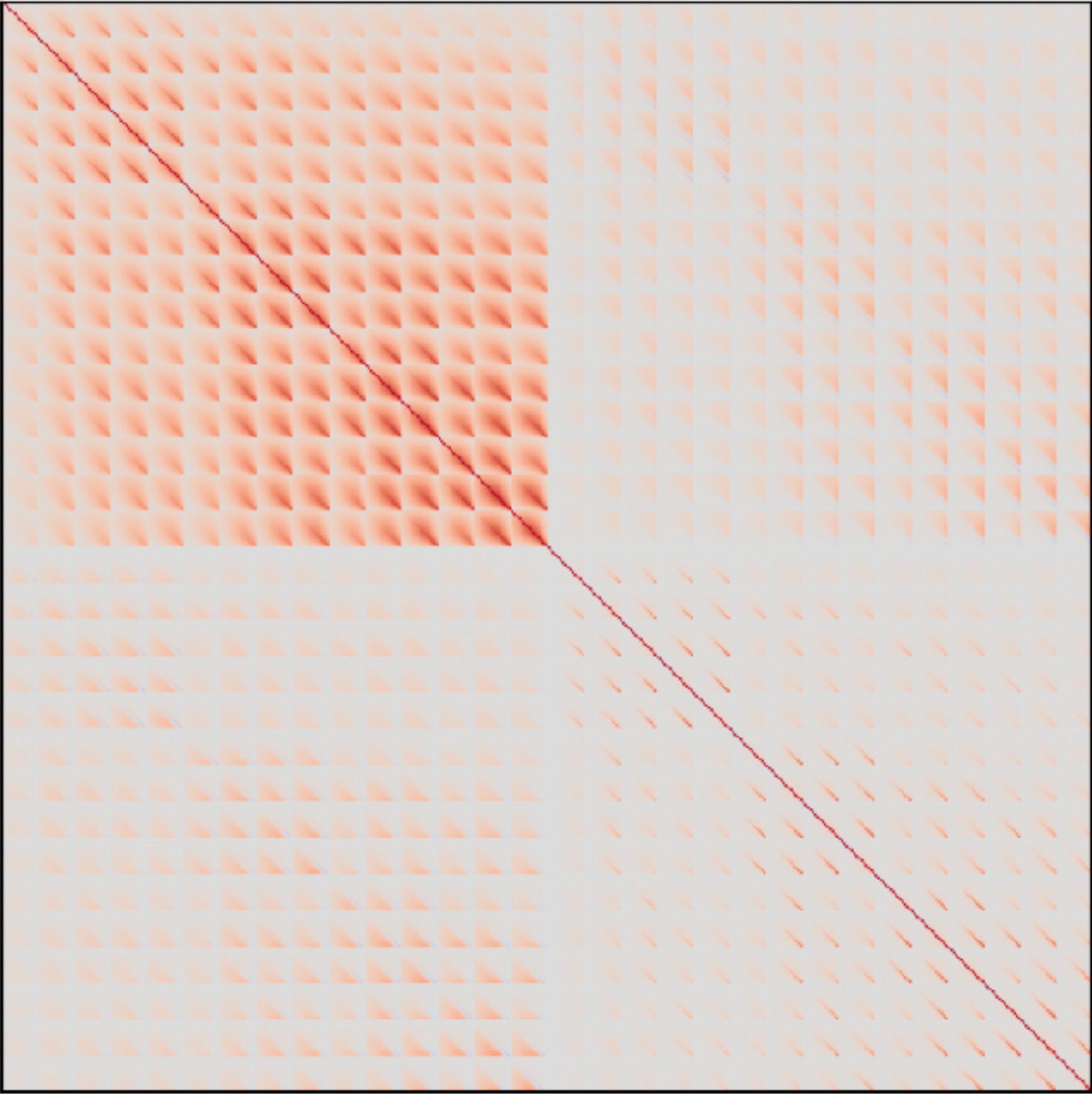


$$\hat{x} \sim \mathcal{G} \left[\hat{x} \mid x[\pi], \Sigma \right]$$

$x[\pi]$



Σ



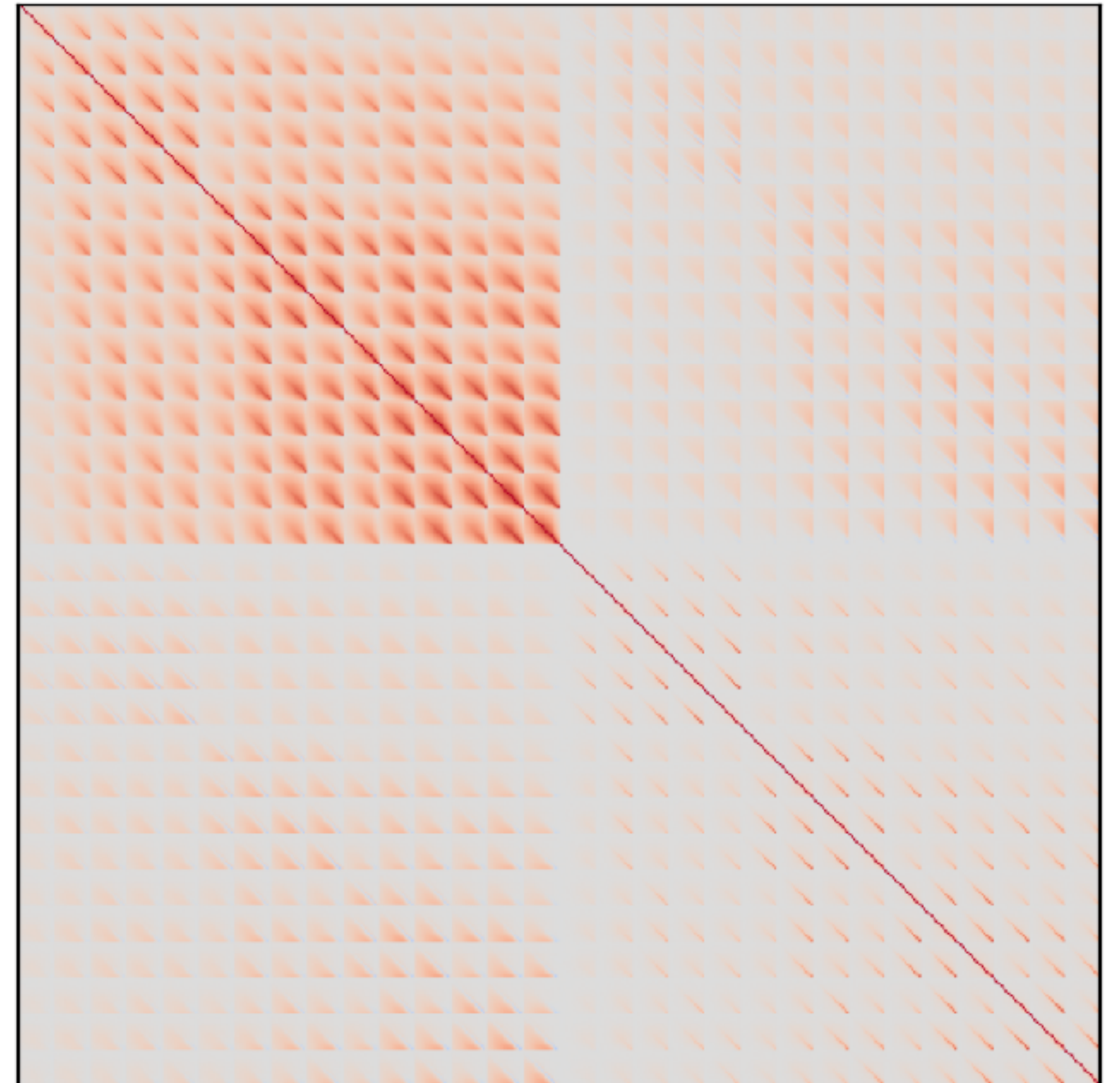
Covariance matrix Σ

- MLE estimator

$$\hat{\Sigma} = \frac{1}{n_s - 1} \sum_{i=1}^{n_s} (x_i - \bar{x})(x_i - \bar{x})^T$$

- Sampling distribution $p(\hat{\Sigma} | \Sigma) = \mathcal{W}[\hat{\Sigma} | \Sigma, \nu]$

Σ



Where do we obtain $\hat{\Sigma} \approx \Sigma$ from?

- Nature doesn't give us Σ , there are 3 ways to obtain an estimator $\hat{\Sigma}$

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Generate simulations of the data, their scatter estimates the covariance.

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2. Internal covariance estimation

Subsets of the data probe the scatter in the data, the covariance.

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1. Covariance estimation from simulated data

Generate simulations of the data, their scatter estimates the covariance.

2. Internal covariance estimation

Subsets of the data probe the scatter in the data, the covariance.

3. Theoretical covariance model

Derive a theory-based model for the covariance from first principles.

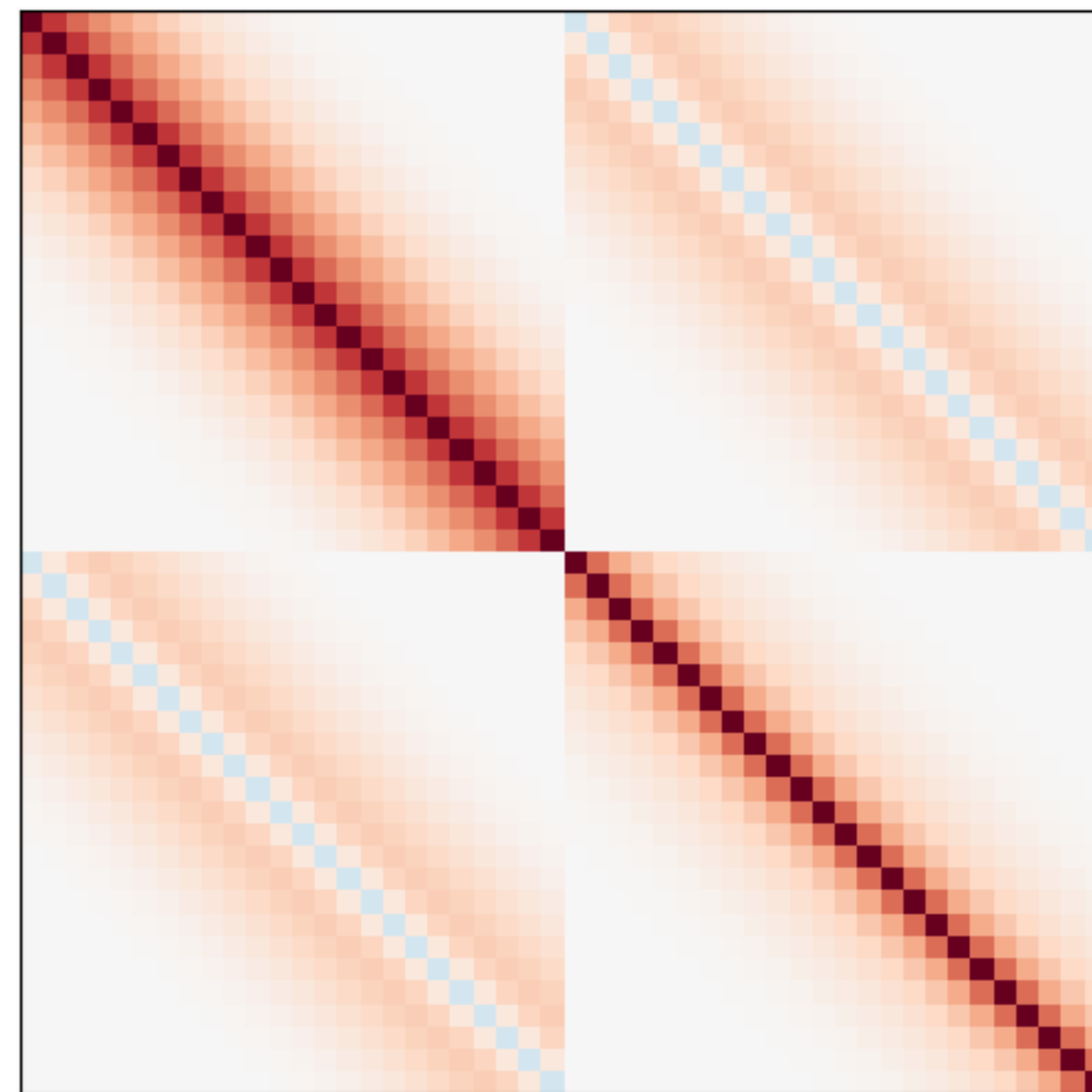
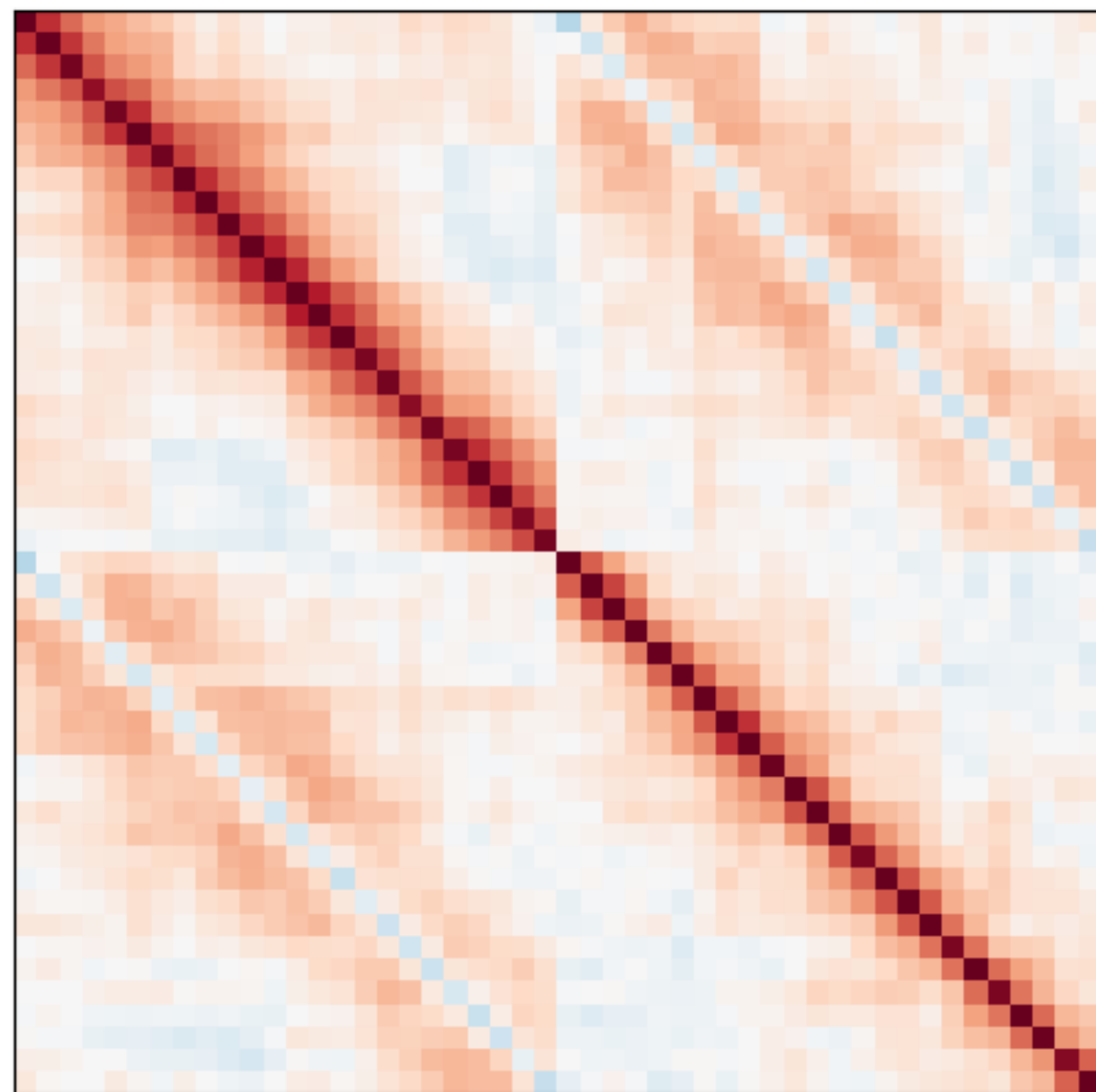
Inverting simulation-estimated $\hat{\Sigma} \rightarrow \Psi$

- Let's say we have a set of n_s simulations $\mathcal{X} = \{x_i, \dots, x_{n_s}\} \sim \mathcal{G}[x | x[\pi], \Sigma]$.
- We then use the MLE $\hat{\Sigma} = \frac{1}{n_s} \sum_{i=1}^{n_s} (x_i - \bar{x})(x_i - \bar{x})^T$.
- What happens when we invert this noisy matrix $\hat{\Sigma}$?

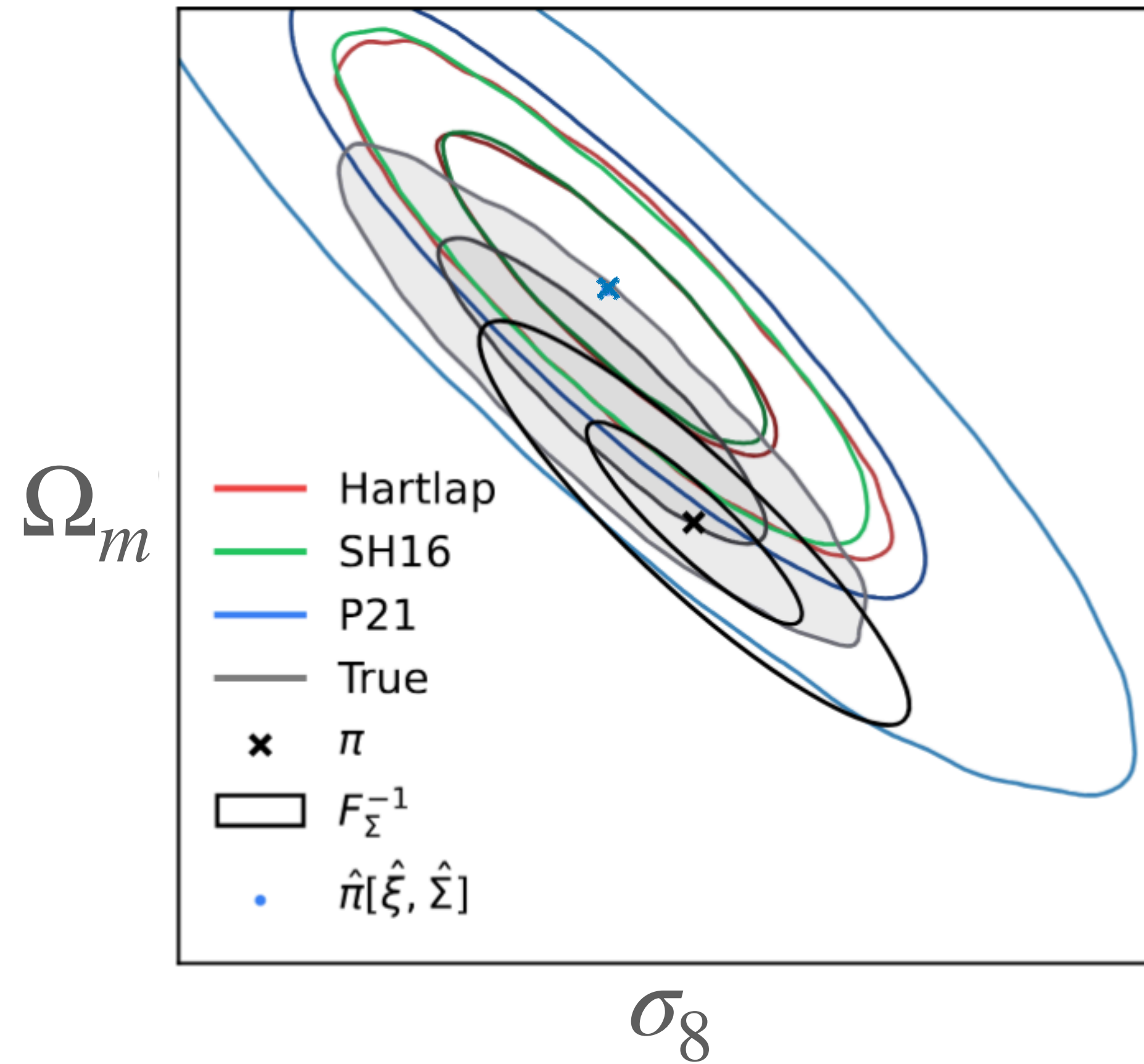
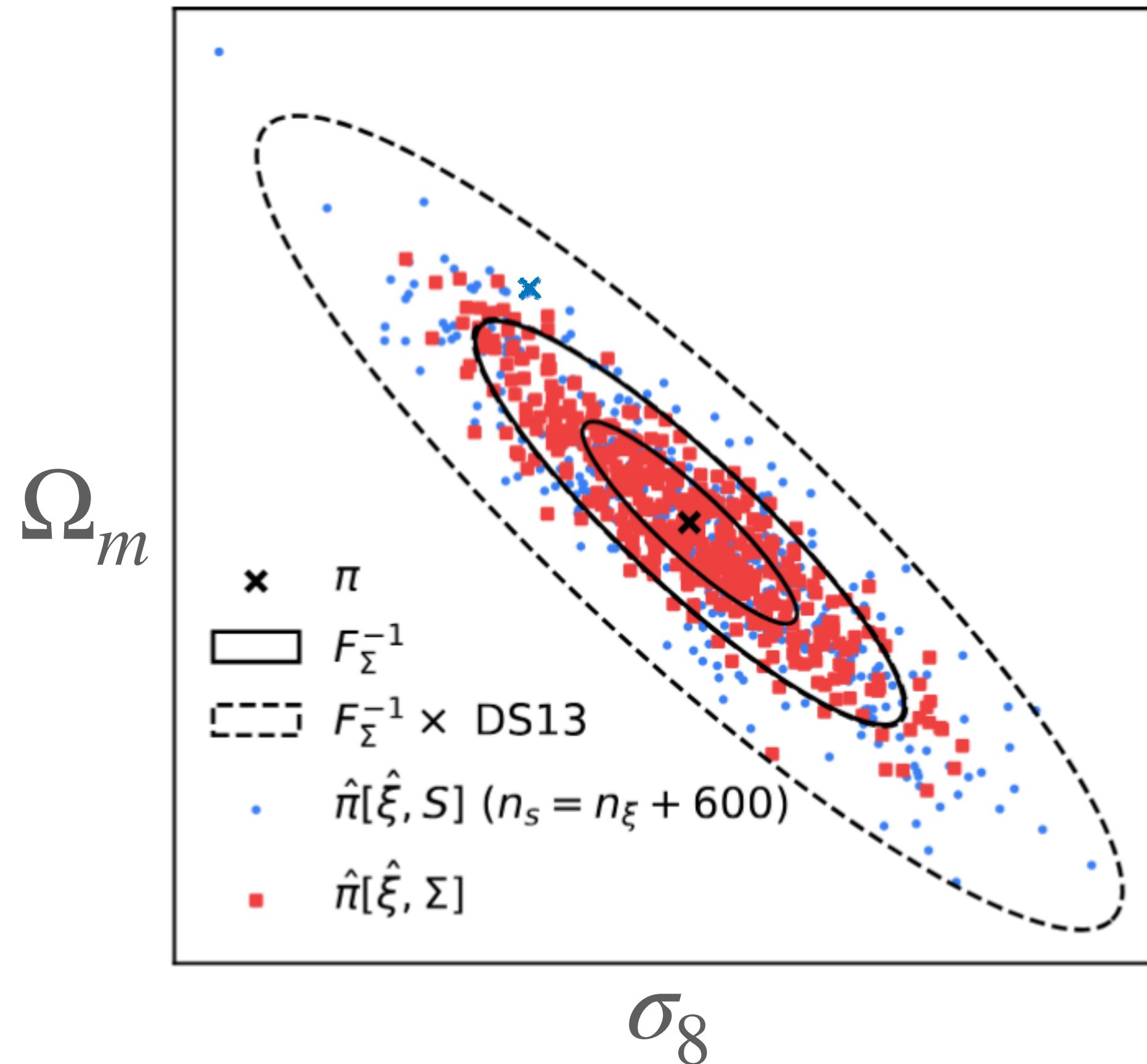
Inverting simulation-estimated $\hat{\Sigma} \rightarrow \Psi$

$$\hat{\Sigma} = \frac{1}{n_s} \sum_{i=1}^{n_s} (x_i - \bar{x})(x_i - \bar{x})^T$$

$n_s = 285$



Problem: estimating π from $\hat{x}$ when $\hat{\Sigma} \neq \Sigma$



Noise in $\hat{\Sigma} \neq \Sigma$ adds scatter to best-fit $\hat{\pi}[\hat{x}, \hat{\Sigma}]$. Not accounted for in Gaussian ansatz.

Distribution of $\hat{\Sigma}$

- If our data x are Gaussian, the distribution of $\hat{\Sigma}$ is given by a Wishart distribution

$$p(\hat{\Sigma}) \propto |\hat{\Sigma}\Sigma^{-1}|^{\frac{n_s - n_x - 2}{2}} \exp\left[-\frac{n_s}{2}\text{Tr}(\hat{\Sigma}\Sigma^{-1})\right]$$

Expectation of $(\hat{\Sigma})^{-1}$

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- The expectation of the $\hat{\Psi} = (\hat{\Sigma})^{-1}$ is a multiple of the true precision

$$\mathbb{E}[\hat{\Psi}] = \frac{n_s - 1}{n_s - n_x - 2} \Sigma^{-1}$$

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This shows us how to unbiased an estimated data covariance! Known as the Kaufman-Hartlap factor.

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Covariance of the covariance

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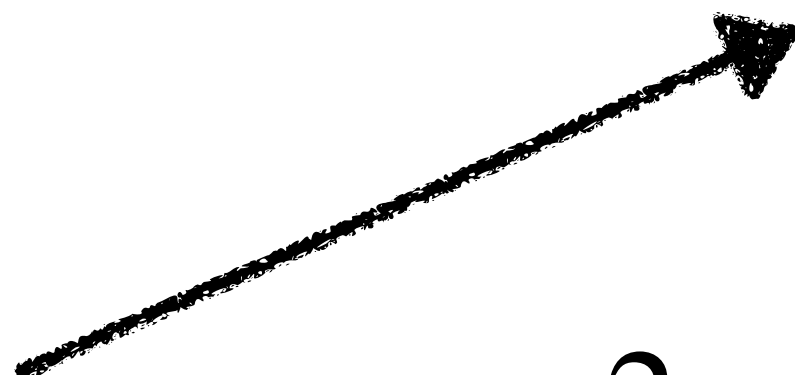
- The covariance of the elements in the estimated covariance is given by

$$\text{Cov}[\hat{\Sigma}_{ij}, \hat{\Sigma}_{kl}] = \frac{1}{n_s - 1} (\Sigma_{ik}\Sigma_{jl} + \Sigma_{il}\Sigma_{jk})$$


The covariance of off-diagonal elements still depends on the diagonal elements!

Inverting simulation-estimated $\hat{\Sigma} \rightarrow \Psi$

- What happens when we invert this noisy matrix $\hat{\Sigma}$?
- $\hat{\Psi}$ depends on the determinant of $\hat{\Sigma}$
 - this is a non-linear combination of $\hat{\Sigma}$ elements
 - this strongly amplifies the noise in the inverse covariance estimate $\hat{\Psi}$...
- This **biases** the estimator $\hat{\Psi}$!

$$\text{Var}[\hat{\Psi}_{ii}] = \frac{2}{n_s - n_x - 4} \Psi_{ii}^2$$


$p(x \mid \pi, \hat{\Psi})$ is a non-linear function of $\hat{\Psi}$

- We are interested in the likelihood $p(x \mid \pi)$, but we're approximating it with the use of $\hat{\Psi}$
- Our likelihood is biased, because $p(x \mid \pi)$ is non-linear in $\hat{\Psi}$

$$\mathbb{E} [p(x \mid \pi, \hat{\Psi})] \neq \mathbb{E} [p(x \mid \pi, \mathbb{E}[\hat{\Psi}])]$$

- This means our likelihood is incorrect...

$$\mathbb{E} [p(x \mid \pi, \hat{\Psi})] \neq \mathbb{E} [p(x \mid \pi, \Sigma^{-1})]$$

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$$\mathbb{E} [p(x \mid \pi,$$

- This means our likelihood is

$$\mathbb{E} [p(x \mid \pi, \hat{\Psi})] \neq \mathbb{E} [p(x \mid \pi, \Sigma^{-1})]$$

We can estimate this bias by the *Percival Correction* (applied to our posterior covariance)

$$P(n_s, n_x, n_\pi) \approx 1 + \frac{n_\pi + 1}{n_s - n_x}$$

Maximum-likelihood estimators $\hat{\pi}$

- Let $\mu = x[\pi]$ so we have $p(x | \pi) = p(x | x[\pi], \pi)$ in general.
- Deriving this equation, showing its covariance...

$$\hat{\pi} = \pi + M[\hat{\Psi}^{-1}] (\hat{x} - x[\pi])$$

Cov $[\hat{\pi}]$

- Let's derive the covariance of this MLE
- It equals the Fisher information at π , $F[\pi]$. What does this mean?

$$\hat{\pi} = \pi + M[\hat{\Psi}^{-1}] (\hat{x} - x[\pi])$$

Best-fit $\hat{\pi}$ with estimated covariance $\hat{\Sigma} \neq \Sigma$

- Estimating **best-fit** $\hat{\pi}$ from $\hat{x}$ with an **estimated covariance** $\hat{\Sigma}$

$$\hat{x} \sim \mathcal{G}[\hat{x} | x[\pi], \Sigma]$$



$$\hat{\pi} = \pi + M[\hat{\Psi}^{-1}] (\hat{x} - x[\pi])$$

$$\hat{\Psi}^{-1}$$

$$\Sigma^{-1}$$

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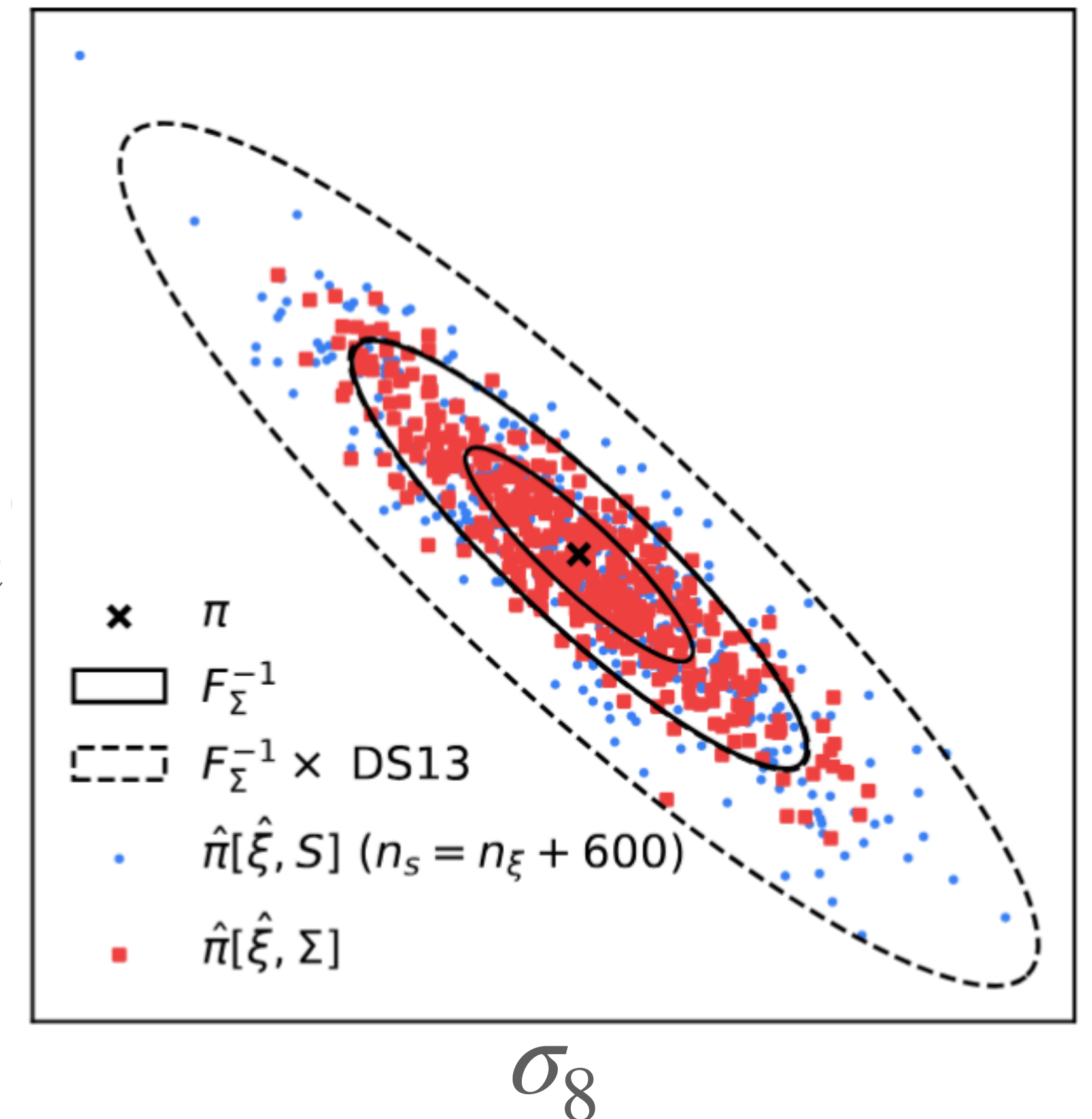


$$\hat{\pi} = \pi + M[\hat{\Psi}^{-1}](\hat{x} - x[\pi])$$

$\hat{\Psi}^{-1}$

Σ^{-1}

Ω_m

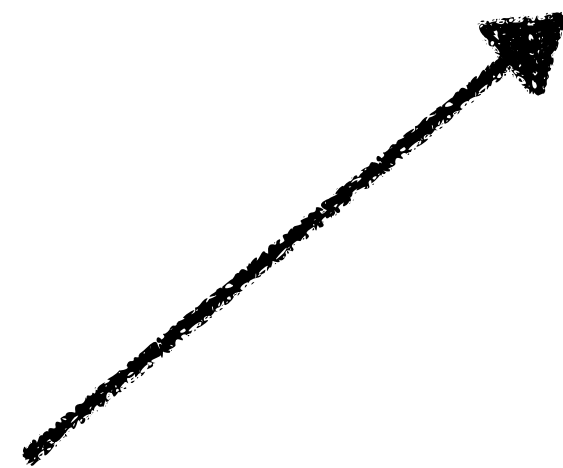


The Dodelson-Schneider Effect

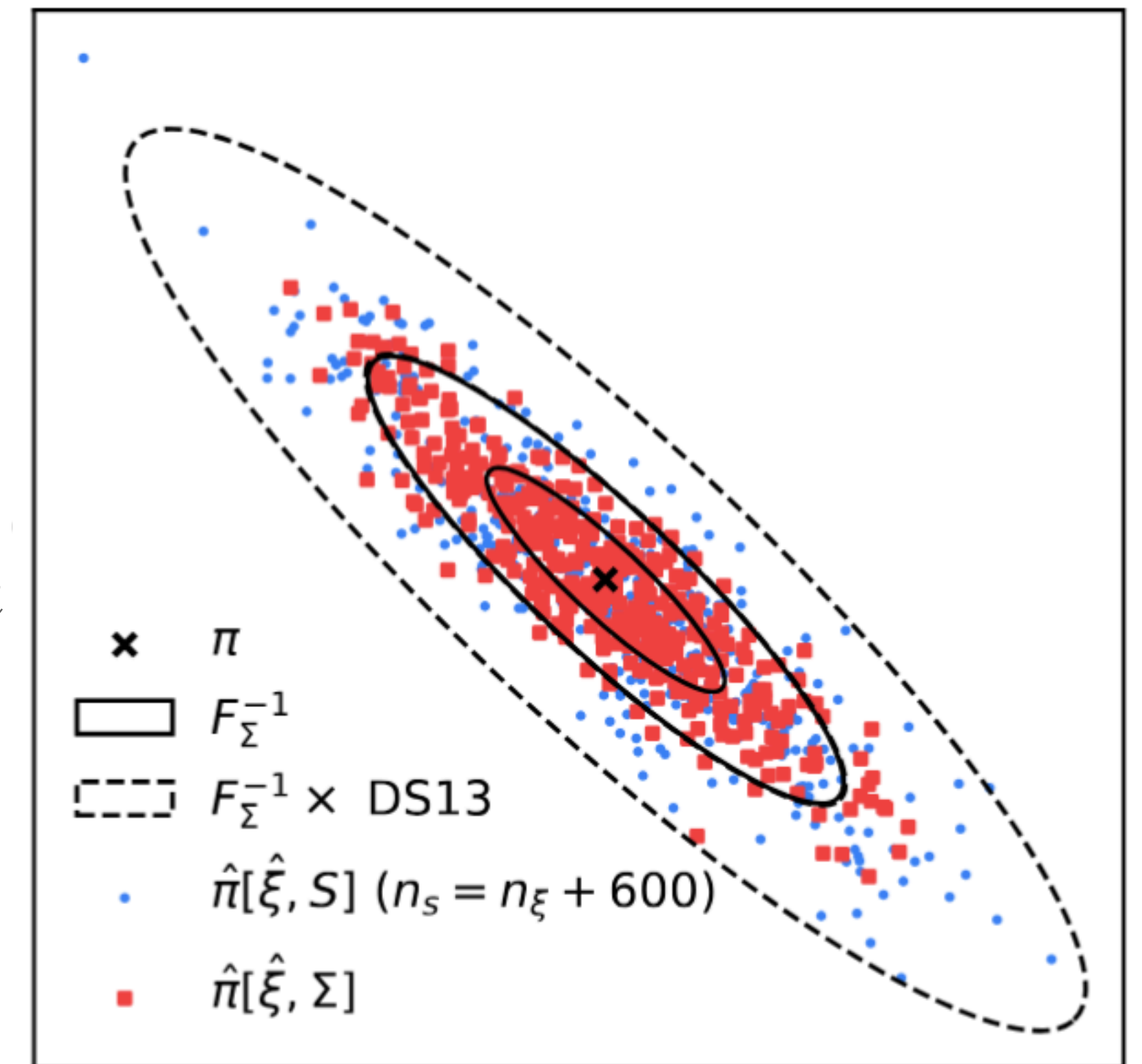
- Estimate the covariance $\hat{\Sigma}$ with n_s simulations

$$\hat{\pi} = \pi + M[\hat{\Psi}^{-1}] (\hat{x} - x[\pi])$$

$$\hat{\Psi}^{-1} = \Sigma^{-1} + \Delta_{\Sigma}$$



Ω_m



σ_8

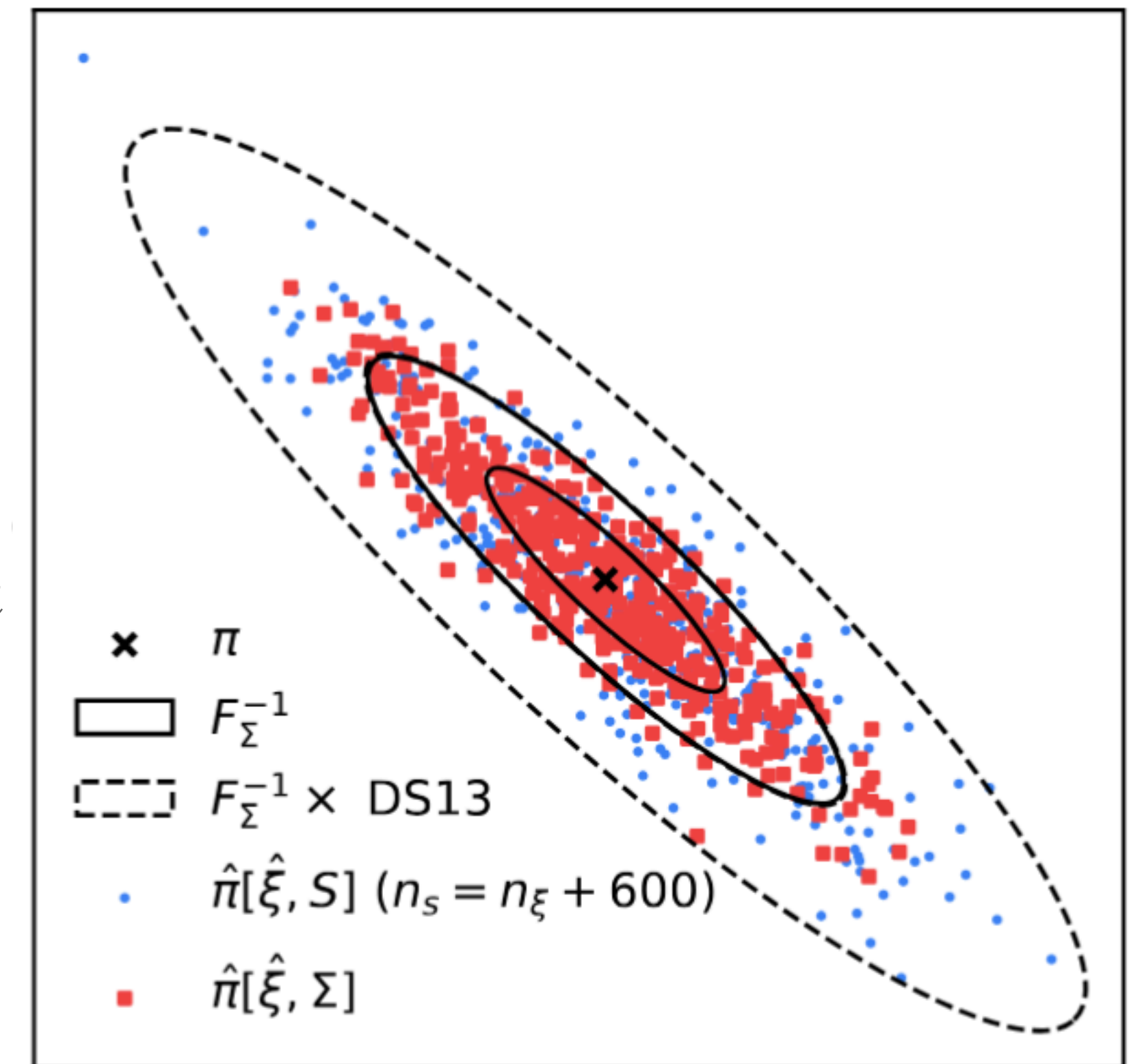
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- Estimate the covariance $\hat{\Sigma}$ with n_s simulations

$$\hat{\pi} = \pi + M[\hat{\Psi}^{-1}](\hat{x} - x[\pi])$$

$$\hat{\Sigma}_{\pi} = \Sigma_{\pi} \left[1 + \frac{n_x - n_{\pi}}{n_s - n_x} \right]$$

Ω_m

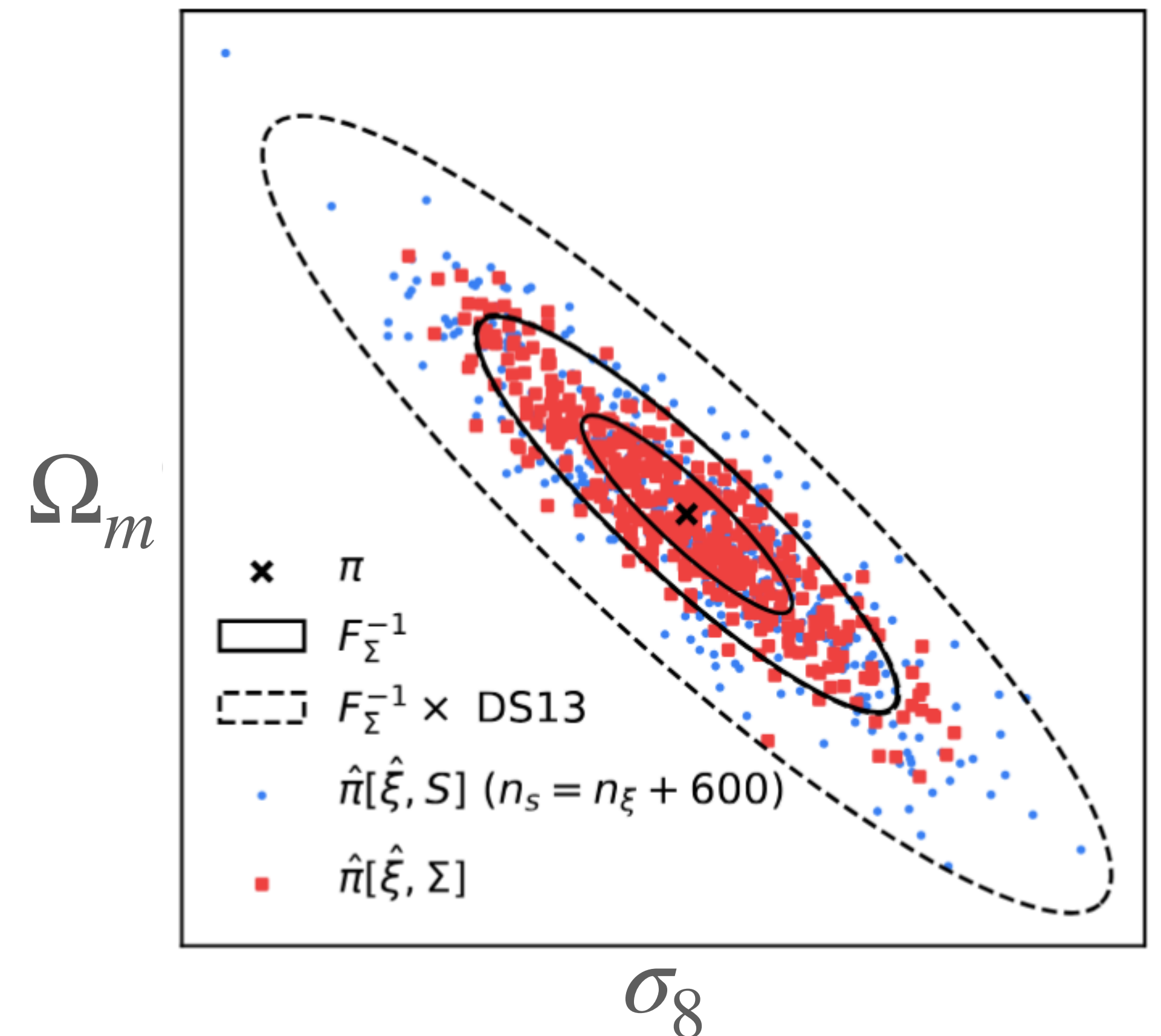


σ_8

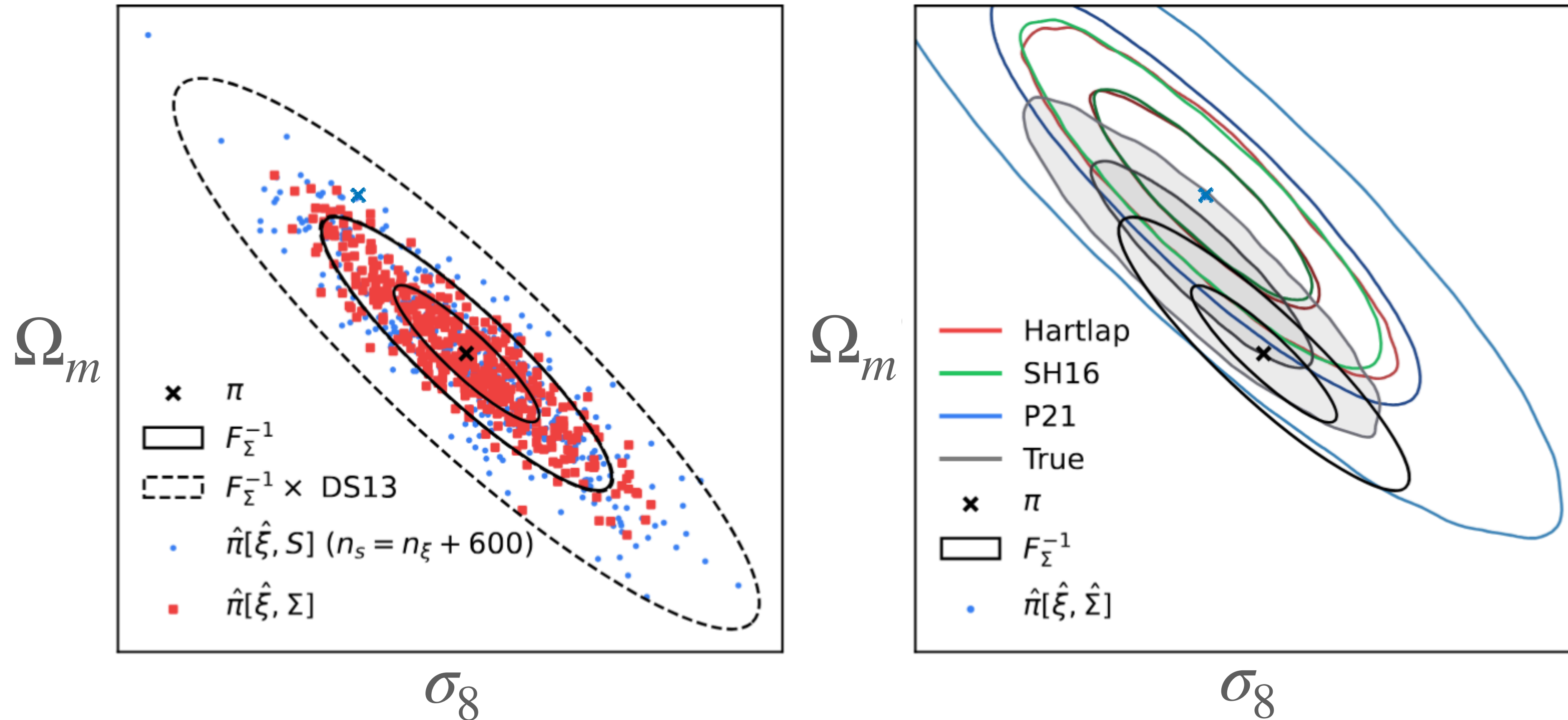
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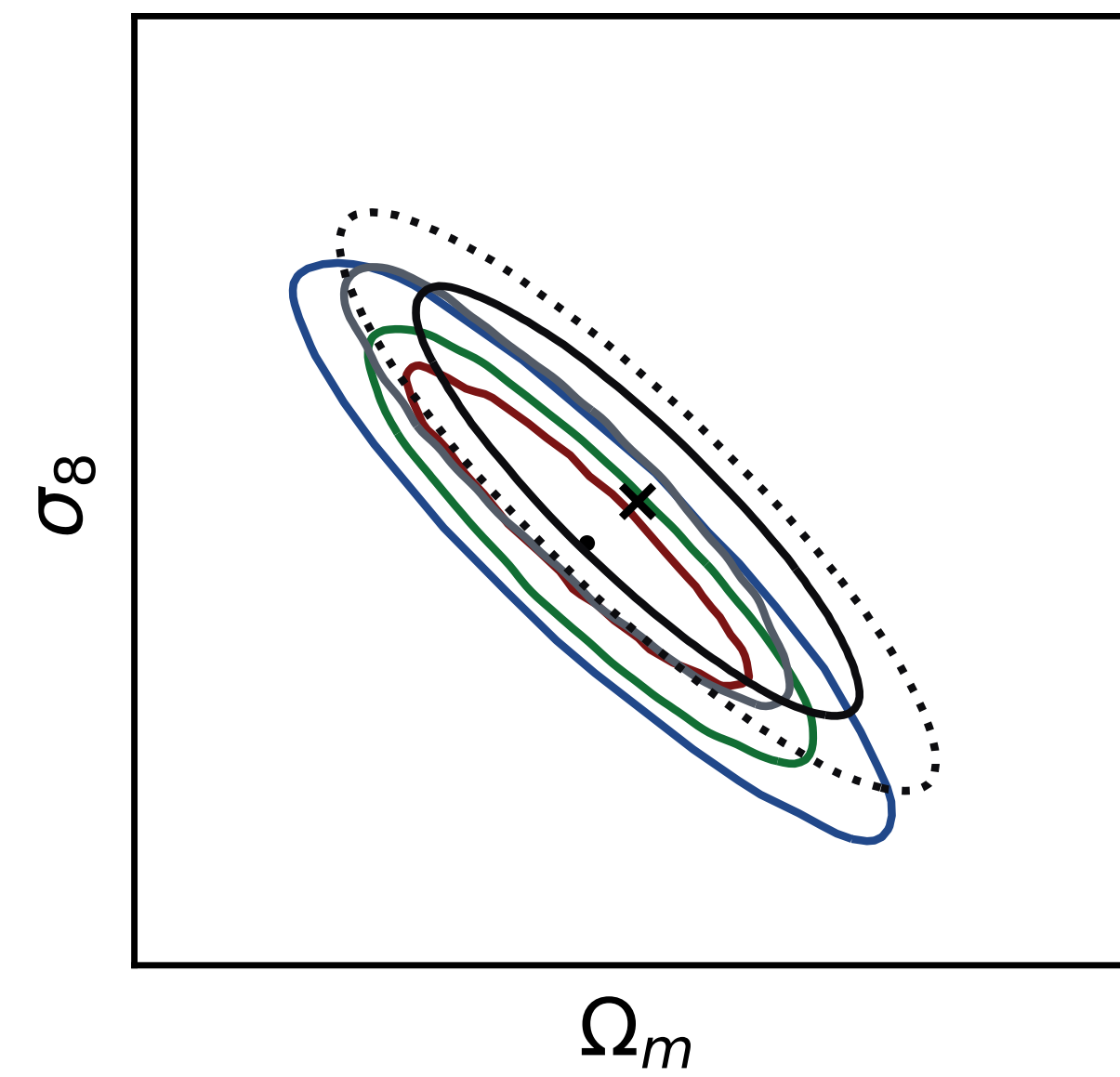
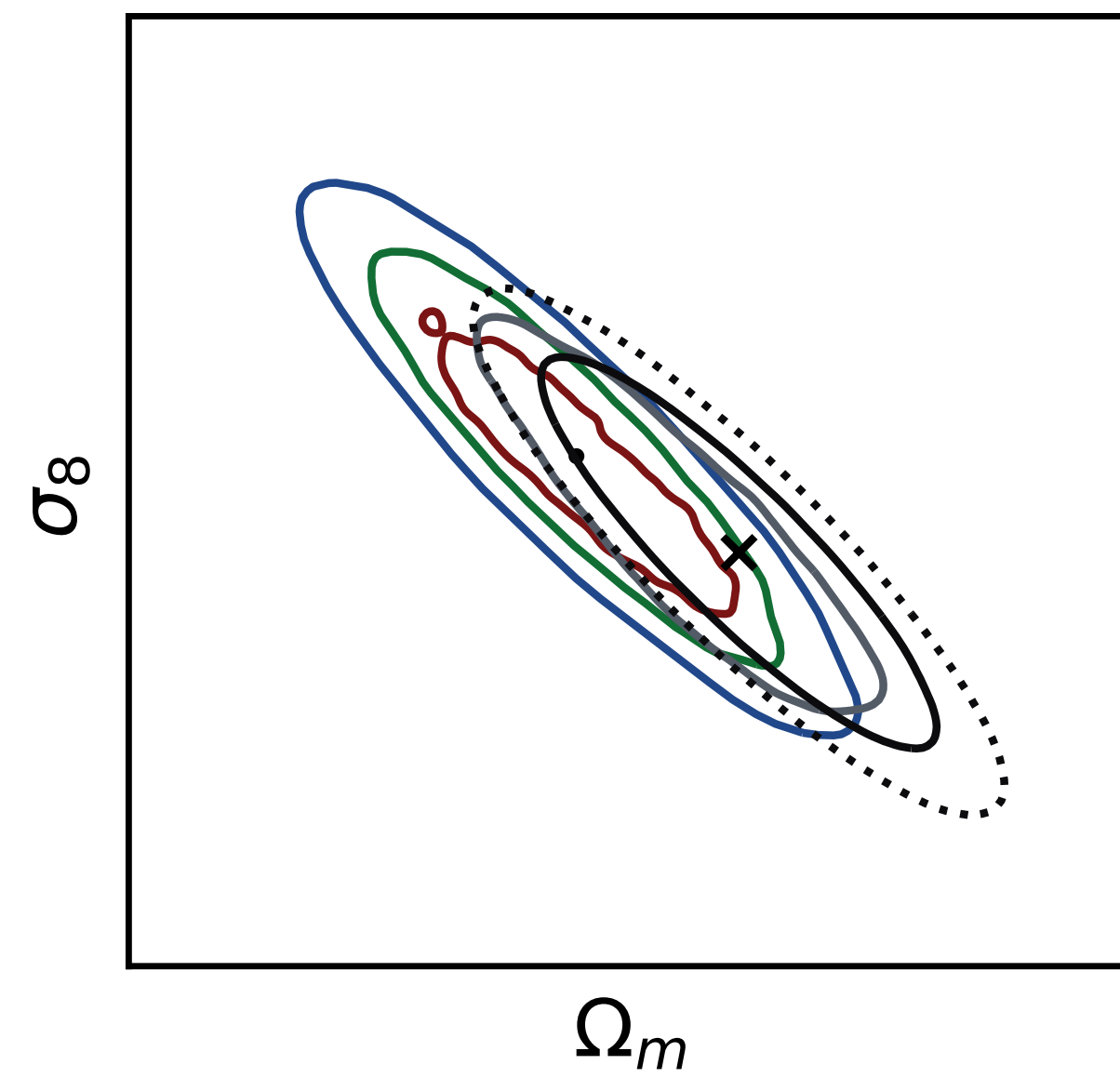
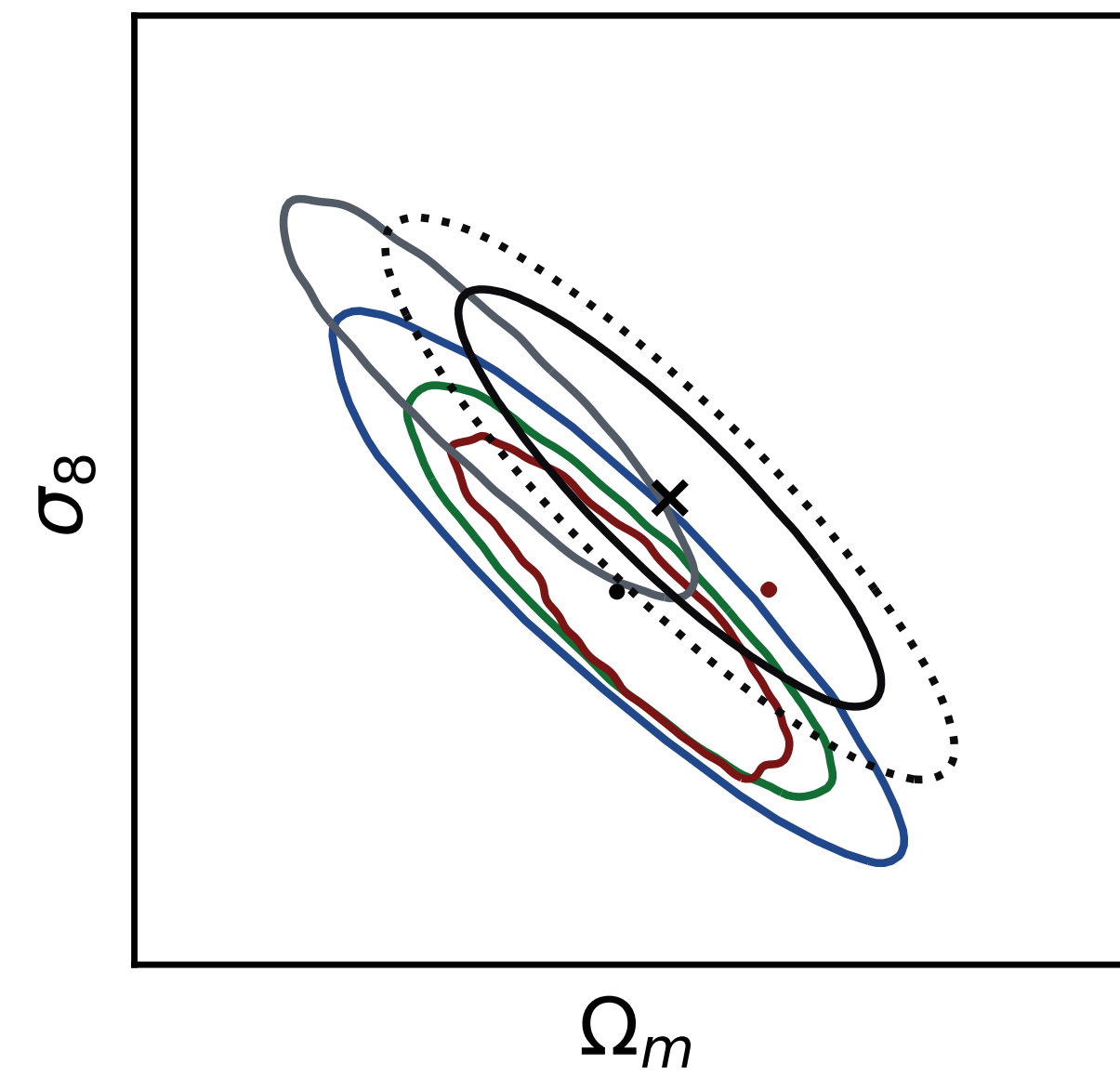
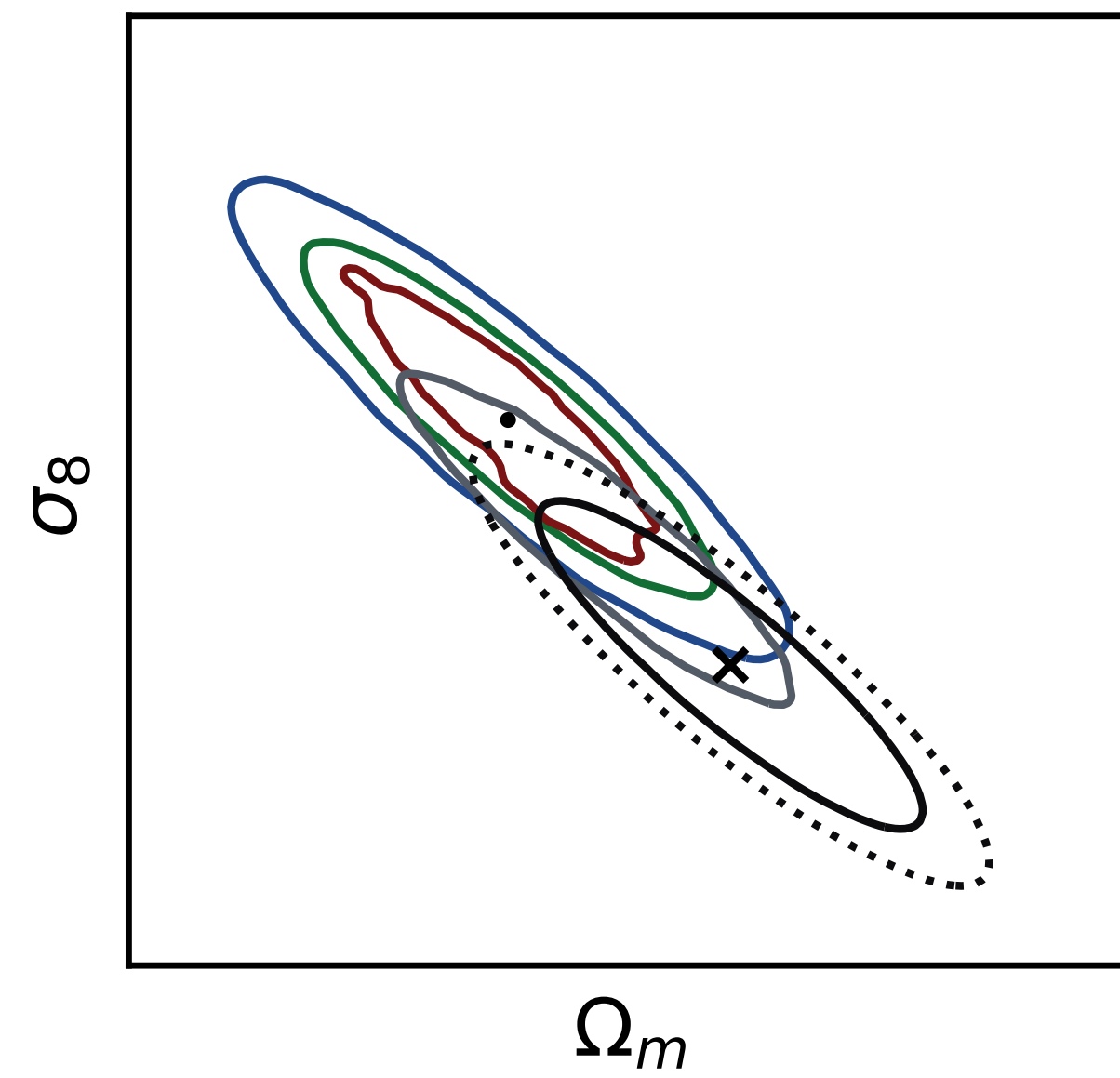
- What does this mean?
- Limits?
- Can we fix it?



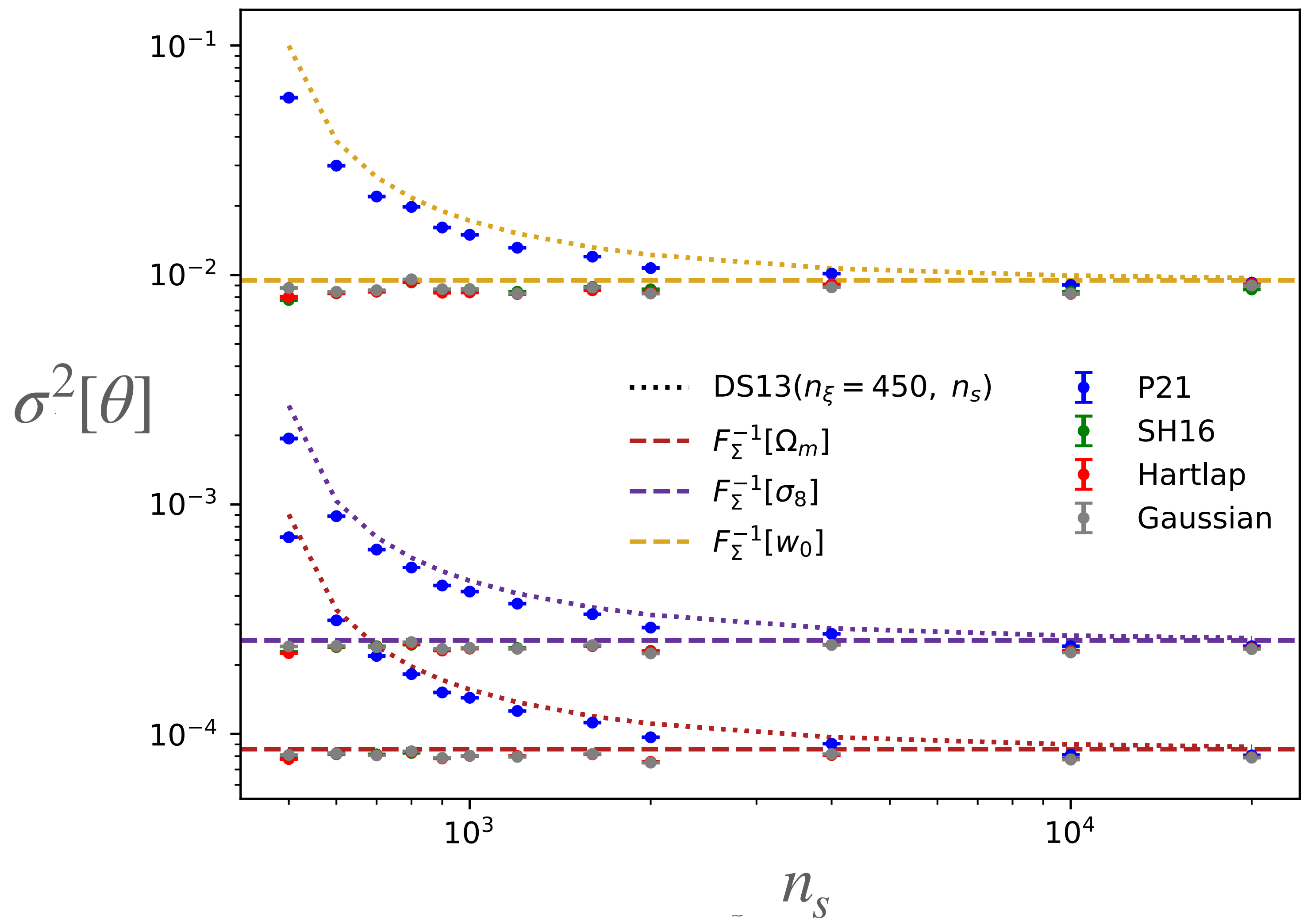
Problem: estimating π from $\hat{x}$ when $\hat{\Sigma} \neq \Sigma$



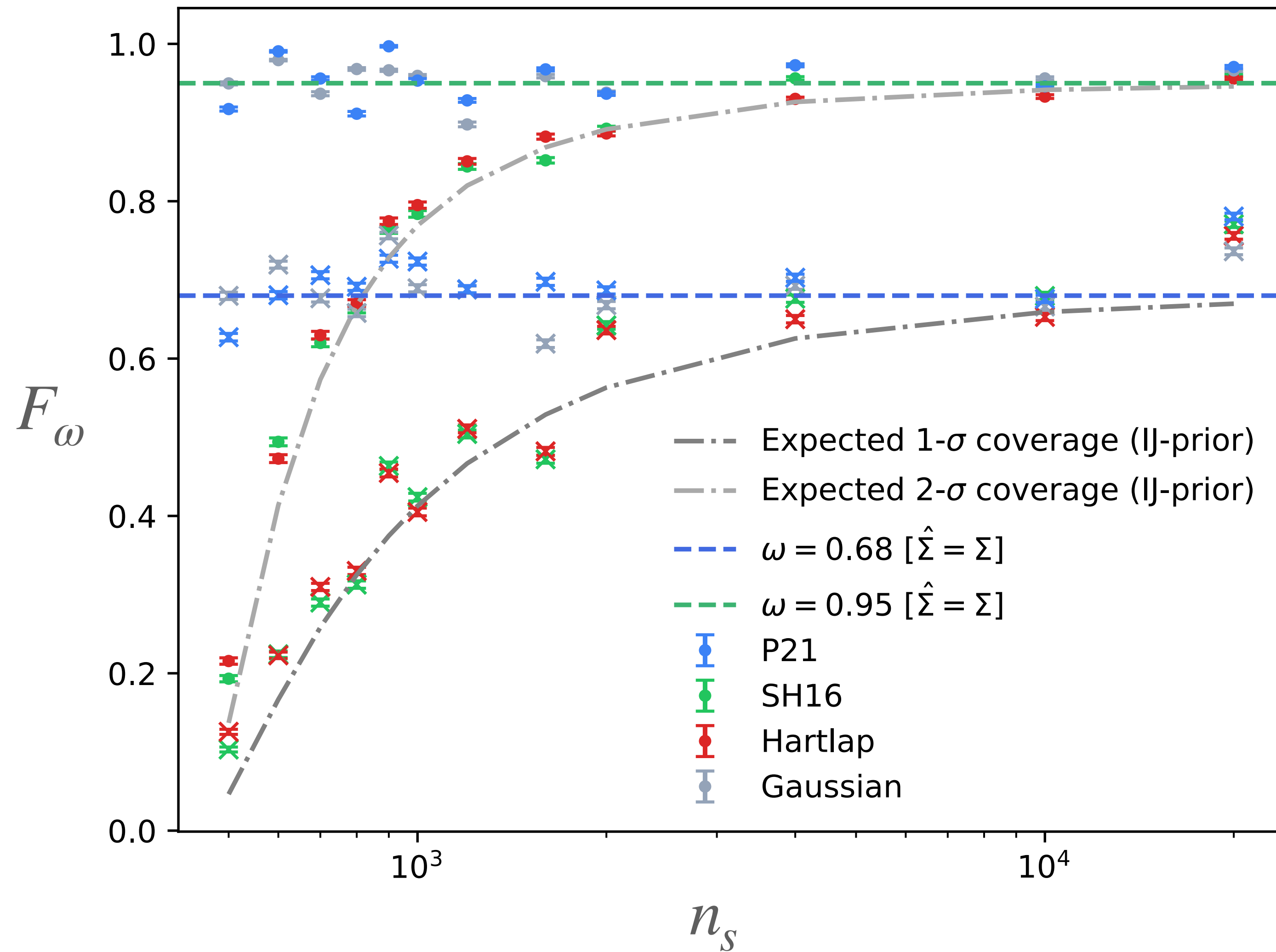
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Posterior widths - Gaussian likelihood analysis



Posterior coverages - Gaussian likelihood analysis



Analytic solution (Percival++21)

- Why did I only talk about Gaussian linear models?
 - First-order approximation to a non-linear model.
 - In high samples limit, approximately holds.

Analytic solution (Percival++21)

- Set a prior on the unknown covariance Σ then marginalise!

$$\begin{aligned} p(\pi | x, \hat{\Sigma}) &= \int d\Sigma \, p(\pi, \Sigma | x, \hat{\Sigma}) \\ &= \int d\Sigma \, p(x | \pi, \Sigma) p(\Sigma | \hat{\Sigma}) p(\pi) \end{aligned}$$

Analytic solution (Percival++21)

- Set a prior on the unknown covariance Σ then marginalise!
- Find parameterisation m of this prior $p(\Sigma) \propto |\Sigma|^{-\frac{m}{2}}$ by *matching* the scatter of the MLE with the expected width of the posterior

$$\langle \delta \hat{\pi} \delta \hat{\pi}^T \rangle_{x, \hat{\Sigma}} \approx \langle \delta \hat{\pi} \delta \hat{\pi}^T \rangle_{\pi|x, x}$$

- This gives the blue contours in our previous plots (see how this behaves!)

Summary

- The covariance Σ is a high-dimensional (n_x^2) object that is critical for $p(x | \pi)$
- There are many ways to estimate Σ , but even when we have accurate simulations, there is a catastrophic effect on our constraints (for realistic inference setups) due to the noisy estimate $\hat{\Sigma}$...
- This causes a scatter in the location and width of the parameter contours (the former being a worsened effect)
- We can account for the effect using the Dodelson-Schneider and Percival corrections, as well as a Frequentist-matching prior (Percival++21).
 - That said, these solutions are only accurate for Gaussian data x and linear models $x[\pi]$

Write down something you learned today and
something you need to hear more about next lecture

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